

Data Tasks: Collection & Preprocessing

Time-Aware Explainable ML for Early At-Risk Student Prediction on OULAD

Group 1 – DSP391m

FPT University

Data Science Capstone (DSP391m) ■ Report 2

Table of Contents

- 1 Required Data for the Study
- 2 Data Collection Method
- 3 Data Cleaning Strategy
- 4 Exploratory Data Analysis
- 5 Data Transformation & Standardisation
- 6 Train/Test Split & Leakage Control

Outline

- 1 Required Data for the Study
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Data Requirements

Requirement	Why it is required	OULAD source
Binary at-risk label	Prediction target — used by all three RQs	<code>studentInfo.final_result</code>
Time-stamped engagement	Time-aware checkpoint features (RQ1/RQ2)	<code>studentVle (~10.6M)</code> , <code>vle</code>
Academic performance	Early score signals up to each checkpoint	<code>studentAssessment</code> , <code>assessments</code>
Demographic context	Baseline covariates + fairness analysis	<code>studentInfo</code> , <code>studentRegistration</code>
Course timeline	Convert event dates to progress %	<code>courses.module_presentation_1</code>

Three feature groups on the key (`id_student`, `code_module`, `code_presentation`): Demographic ▪ Engagement (VLE)
▪ Performance (`assessments`).

OULAD at a Glance

Scale of the dataset

- **32,593** student–module–presentation records
- **28,785** unique students
- **22** module-presentations, **7** tables
- **10,655,280** VLE click events
- **~173,912** assessment submissions
- Daily resolution (2013–2014)

The Open University (UK); CC-BY 4.0, anonymised at source.

Table	Role	Time?
studentInfo	Outcome label + demographics	no
studentRegistration	Registration / withdrawal dates	part
studentVle	Clickstream (~10.6M rows)	yes
vle	Maps site → activity type	no
studentAssessment	Submissions + scores	yes
assessments	Type, weight, deadline	part
courses	Presentation length (days)	no

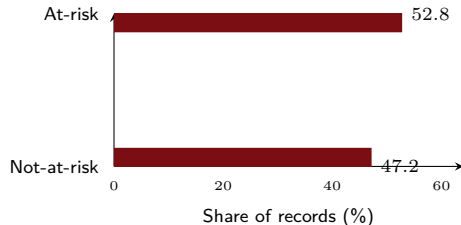
studentVle.date and studentAssessment.date_submitted are the foundation of the time-aware design.

Target Variable & Class Balance

final_result	Group	Label
Distinction	Pass	not-at-risk (0)
Pass	Pass	not-at-risk (0)
Fail	At-risk	at-risk (1)
Withdrawn	At-risk	at-risk (1)

`at_risk = final_result ∈ {Fail, Withdrawn}`

- Label **fixed** across all six checkpoints.
- Imbalance is **mild** ($\approx 52.8\%$ at-risk).
- Metrics: **PR-AUC** + **recall** on the at-risk class.



17,208 at-risk vs. 15,385 not-at-risk.

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Choosing a Data Source

Four source categories were weighed; public / secondary data wins on the criteria that matter here.

Criterion	Internal DB	API	Scraping	Public
Immediate availability	Low	Medium	Medium	High
Already anonymised	No	No	No	Yes
Reproducibility	Low	Low	Very low	High
Comparability w/ prior work	Low	Low	Low	High
Ethical / legal complexity	High	Medium	High	Low (CC)

*OULAD is **secondary data** — collected and anonymised by The Open University, then released: we trade collection-design control for reproducibility and comparability.*

Why OULAD — and License & Ethics

Selection criteria $\Rightarrow$ OULAD fits

- Publicly downloadable — reproducible by anyone.
- Three feature groups present — demographics, VLE, assessments.
- Labelled target — `final_result` $\rightarrow$ `at_risk`.
- Used by the base studies (Adnan 2021; Tomasevic 2020).
- Static snapshot — identical file for everyone.

License & ethics

- **CC-BY 4.0** — reuse with attribution only.
- Anonymised at source: no IDs; region-level; banded age & IMD.
- No re-identification; no external data combined.
- Raw CSVs read-only + md5 verification.
- Cite Kuzilek et al. (2017), *Scientific Data*.

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Cleaning — Problems, Methods, Outcomes

Problem	Method	Outcome
Duplicates	<code>drop_duplicates</code> on the composite key	0 duplicate keys
Inconsistency	<code>str.strip()</code> on categoricals (reversible)	Stable cardinalities
Missing values	Variable-specific imputation, fit on train	0 NaN after cleaning
Outliers	IQR $\rightarrow$ log1p / winsorise; no rows dropped	Skew tamed, records kept
Temporal	<code>cut_at_checkpoint</code> removes future rows	No look-ahead leakage

Order: structural cleaning $\rightarrow$ missing $\rightarrow$ outliers $\rightarrow$ encode/scale — every learned parameter comes from the training fold only.

Missing Values & Outliers

Variable	#Miss	Strategy
imd_band	1,111	→ “Unknown” (rank-0)
score*_to_date	gaps	→ 0 + not_submitted
date_registration	45	→ train median
date_unregistration	22,521	dropped — not a feature

A missing score is signal (“nothing submitted yet”) — made explicit by the `not_submitted` flag.

Outliers — IQR rule, train thresholds

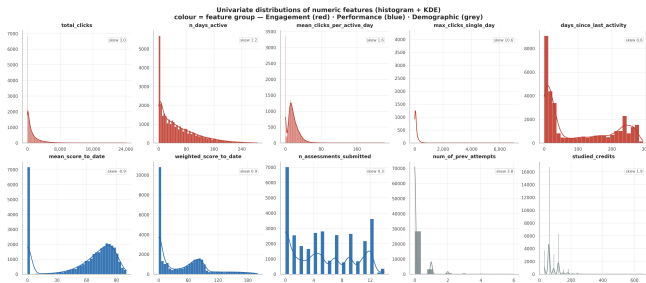
- **log1p** — right-skewed clickstream (`total_clicks` skew ≈ 3.0 ; `max_clicks_single_day` ≈ 10.6).
- **winsorise 1%** — `studied_credits`, `num_of_prev_attempts`, `weighted_score`.
- **none** — naturally bounded (`mean_score`, `n_assessments`).

No record is deleted — ranks preserved.

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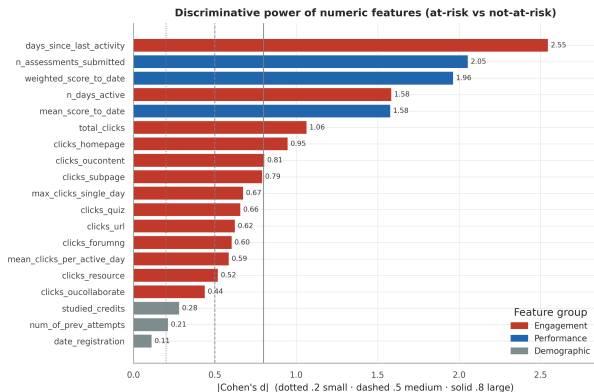
EDA — Univariate Distributions



Histogram + KDE per numeric feature

- Most clickstream features are **strongly right-skewed**: `clicks_resource` skew **34.7**, kurtosis $\approx 2,125$.
- Heavy tails are **real** outliers (super-active students) $\Rightarrow$ `log1p`, not deletion.
- `days_since_last_activity` is **bimodal** — the disengagement signature.
- Non-normal $\Rightarrow$ non-parametric **Mann–Whitney** tests downstream.

EDA — Numeric Features vs At-Risk (Cohen's d)

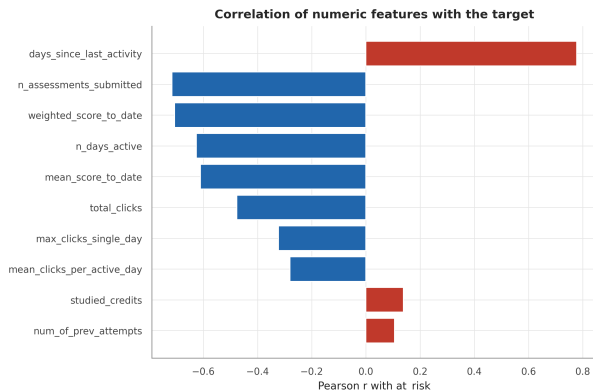


Effect size + Mann–Whitney (BH-corrected)

- Strongest: `days_since_last_activity` $d = 2.55$ (171 vs 14 days), `n_assessments` 2.05 (2.3 vs 8.6), `weighted_score` 1.96.
- All 19/19 numeric features significant ($q < 0.05$) — at $n \approx 32k$, rank by effect size, not p .

Behaviour & performance, not demographics, drive risk.

EDA — Correlation & Leakage Check



Pearson + Spearman; correlation with the label

- Top $|r|$ with label: `days_idle` **+0.78**, `n_assessments` **-0.72**, `weighted_score` **-0.71**.
- 2 multicollinear pairs $|r| \geq 0.8$ (`n_days_active`-`total_clicks` 0.84) — flagged for SHAP.
- **No feature** $|r| \geq 0.95 \Rightarrow$ no leakage; demographic Cramér's $V \leq 0.15$.

Independent methods (d & r) agree $\Rightarrow$ trustworthy signal, no leakage.

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Variable Typing & Encoding

28 input features → five types, each routed to the right transformer in one `ColumnTransformer`.

Type	#	Examples	Encoder / scaler
Numeric	19	clicks, scores, studied_credits	StandardScaler
Ordinal	3	highest_education, imd_band, age_band	OrdinalEncoder (fixed order)
Nominal	3	region, code_module, code_presentation	OneHotEncoder
Binary	2	gender, disability	custom BinaryEncoder
Indicator	1	not_submitted	passthrough

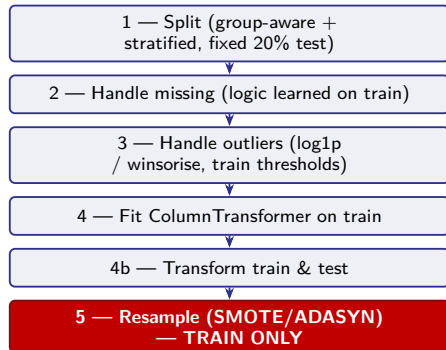
Ordinal keeps the rank; one-hot avoids a false order; `drop=None` keeps every category visible for SHAP / LIME.

Standardisation: Before → After

Component	Before	After
Labels	Dataset-specific class names	Binary: at-risk = 1, not = 0
Categorical features	Text / category values	Encoded numeric (ordinal/one-hot/binary)
Numeric features	Raw counts & scores, skewed	log1p/winsorise + z-score scaled
Feature matrix	28 mixed-scale columns	49-column dense, 0 NaN
Names	Raw column names	snake_case, transformer-prefixed

*StandardScaler (mean 0, sd 1) fitted on **train only**; applied to test with `.transform()` — never `.fit_transform()`.*

Preprocessing Sequence — Fit on Train Only



Why the order matters

- Split **first** — nothing learned before the test set is set aside.
- Impute / encode / scale **fit on train**, applied to test.
- Outlier thresholds per fold; **no rows dropped**.
- Resample **train only** — the test set keeps the real distribution.
- Mirrors the time-axis cut (`cut_at_checkpoint`).

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Split Design

Three data properties constrain the split — so the design is grouped, stratified *and* time-consistent.

1. Grouped

A student spans several module-presentations $\Rightarrow$ split by `id_student` to avoid group leakage.

2. Imbalanced

At-risk $\approx 52.8\% \Rightarrow$ *stratify* so the test ratio cannot drift.

3. Time-aware

Six checkpoints share one axis $\Rightarrow$ the test set is *identical* at every checkpoint.

Design: 20% hold-out (fixed once)

+ 5-fold $\times$ 5-seed CV via `StratifiedGroupKFold`. Verified on `master_raw` (32,593 rows), identical across all 6 checkpoints:

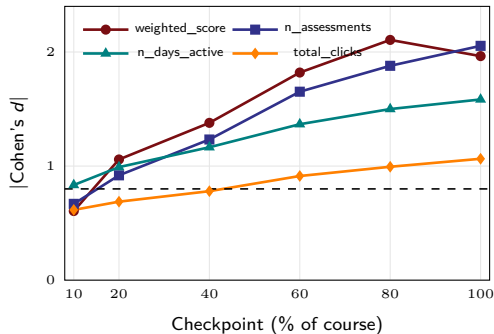
Train rows	Test rows	Test students	At-risk (tr/te)	Overlap
26,104	6,489	5,756	.530 / .520	0

Time-Aware Checkpoints

Features are recomputed at six course-progress checkpoints:

10 ▪ 20 ▪ 40 ▪ 60 ▪ 80 ▪ 100%

- Cutoff day = $\text{round}(\text{length} \times t/100)$. e.g. AAA/2013J (268 d): $t_{10}=27$, $t_{40}=107$, $t_{100}=268$.
- Only rows dated $\leq$ cutoff feed each checkpoint.
- Discrimination **grows with time** — early checkpoints are the hard, valuable ones.



Leakage Prevention — and It Is Tested

Three temporal rules (per checkpoint t)

- 1 Drop assessment submissions with `date_submitted > cutoff`.
- 2 Drop VLE interactions with `date > cutoff` (via `cut_at_checkpoint`).
- 3 Withdrawn-before- t kept & labelled at-risk — low activity is a legitimate early-warning signal, not leakage.

+ **Fit-on-train principle:** every learner (imputers, encoders, scaler, resampler) is fit on the training fold only.

Verified automatically by `tests/test_leakage.py`

19 / 19 tests pass

no record beyond its cutoff ▪ counts non-decreasing in t ▪ $t=100\%$ keeps all ▪ 0 overlap ▪ ratio preserved ▪ train-only impute/winsorise
▪ no-leak feature allow-list

Summary

- 01 Required data** — OULAD supplies label, engagement, performance & demographics at the (student, module, presentation) grain.
- 02 Collection** — a public, anonymised, CC-BY 4.0 benchmark used by the base studies $\Rightarrow$ reproducible & comparable.
- 03 Cleaning** — duplicates, consistency, missing, outliers handled with no record dropped; parameters learned on train.
- 04 Transformation** — 28 features $\rightarrow$ typed encoders $\rightarrow$ a 49-column, leakage-safe, SHAP/LIME-ready matrix.
- 05 Split & leakage** — fixed grouped/stratified 20% test + 5×5 CV; time-aware cuts; 19/19 tests pass.

Thank you

Thank you

Key references

Kuzilek, Hlosta, Zdrahal (2017), *OULAD*, Scientific Data 4:170171 — CC-BY 4.0.
Adnan et al. (2021), *IEEE Access* 9. Tomasevic et al. (2020), *Computers & Education* 143.
Chawla et al. (2002), *SMOTE*, JAIR.